

The Role of In Vitro Directed Chemotherapy in Epithelial Ovarian Cancer

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Epithelial ovarian cancer (EOC) continues to be the most lethal gynecologic malignancy. Efforts to personalize chemotherapy treatments by utilizing in vitro tumor assays to predict chemotherapeutic response have been tested in both the primary and recurrent treatment setting. To date, several retrospective studies have suggested improved response rates to predicted chemotherapeutic agents. However, a prospective, controlled trial merely found equivalence between in vitro prediction and empirical treatment selection. This review summarizes the current data regarding in vitro directed chemotherapy in EOC.

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Epithelial ovarian cancer (EOC) claims the lives of more US women annually than any other gynecologic malignancy.¹ The poor prognosis observed with EOC is related to 2 well-recognized clinical facts: first, a majority of women present with late-stage disease, and, second, tumors that initially respond to chemotherapy inevitably manifest resistance. There has been a relative lack of significant advancement in the treatment of EOC during the past 15 years despite investigations into more aggressive chemotherapeutic approaches.² One notable exception is the resurgence of intraperitoneal (IP) chemotherapy.³

For primary EOC, the chemotherapy regimen that is universally accepted as optimal treatment is the combination of a platinum and taxane, either intravenous or IP. When this regime is combined with optimal cytoreductive surgery, 80% of women can expect to achieve clinical remission.⁴ However, the relapse rate remains high, accounting for the dismal 20% 5-year overall survival (OS) rate for

women with extraovarian EOC at the time of initial diagnosis.⁵

Recurrent EOC is generally divided into 2 groups: platinum sensitive and platinum resistant. Women with tumor recurrence more than 6 months from the time of completion of primary therapy are considered platinum sensitive. In this situation, retreatment with a platinum-based regimen is the typical approach.^{6,7} Women who develop recurrent disease less than 6 months from completion of primary treatment (ie, platinum resistant) face less well-defined treatment options. The US Food and Drug

and lapatinib response in breast cancer, *ras* mutation status determining response to cetuximab in gastrointestinal tumors, and *bcr-abl* protein targeting in chronic myelogenous leukemia.¹⁰⁻¹² Although no such target has been identified in women with EOC, an increasingly sophisticated understanding of the genetic and epigenetic underpinnings of this disease provides optimism for similar interventions in the future.

In the absence of a particular molecular target, phenotypic categorization of tumors in general, and EOC specifically, has the potential to en-

of cell culture, exposure to the test drug, analysis of results, and verification of positive and negative controls. Each available assay is a variation on this theme. Traditionally, these assays are broadly categorized as either sensitivity tests or resistance tests. The distinction between them is subtle and semantic. Chemotherapy sensitivity assays predict agents, or combinations of agents, to which a particular tumor may respond. Chemotherapy resistance assays identify agents to which a particular tumor may not respond. In other words, sensitivity assays more effectively establish the positive predictive value (PPV) and resistance assays more effectively establish the negative predictive value (NPV). Ideally, the clinical scenario should direct which assay type is most appropriate. For example, in the setting of primary EOC, where a majority of women are expected to respond to chemotherapy, a test with a high NPV would be most useful in identifying agents to which response rates would likely be low. In contrast, patients with recurrent EOC are heavily pretreated and the prevalence of chemoresistance is high. In this setting a test with high PPV would be

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Administration (FDA) has approved paclitaxel, pegylated liposomal doxorubicin, gemcitabine, and topotecan among other chemotherapeutic regimens for use in platinum-resistant, recurrent EOC, all of which have a response rate of 20% to 30%.^{8,9} Few women will achieve remission and it is usually not sustainable. The decision making that goes into choosing which agents to use is somewhat empirical, balancing toxicity profiles with the potential for improved disease-free interval and potential survival.

Is there a more scientific approach to choosing a chemotherapy agent which is best for a particular patient? Can we better identify an effective regimen for a specific patient without the wasted time and excessive toxicity associated with empirical treatment? Evidence from other cancer sites (eg, breast, gastrointestinal, leukemia) shows us that targeting inherent tumor vulnerabilities can dramatically increase response rates. Three now classic examples include human epidermal growth factor receptor-2 amplification

and enhance chemotherapy response rates. Several commercially available tests seek to predict a clinical tumor response based on in vitro cell behavior. These tests are collectively known as chemotherapy sensitivity and resistance assays (CSRAs). The concept is straightforward: test each particular patient's tumor and determine the

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specific chemotherapy that will provide maximum response. This review catalogs the various assays available for EOC and reviews the published evidence supporting or dissuading their use in clinical practice.

CSRAs

Elements common to most in vitro chemotherapy prediction assays include: tumor sampling, establishment

more useful in identifying agents to which there is a better chance of clinical response.

There are several reported methods for performing the chemotherapy sensitivity assay in the setting of EOC. In one such assay, tetrazolium (MTT) dye indirectly measures the number of surviving cells following chemotherapy exposure. Once exposed to MTT, enzymes in viable cells use the dye as

a substrate and result in the formation of blue crystals. The color density is then measured and is proportional to the surviving cell population. A second approach utilizes the conversion of nonfluorescent luciferin to its fluorescent form as a measure of the adenosine triphosphate (ATP) content of viable cells following drug exposure.

A drug-resistance assay commonly cited in the literature, which is also commercially available, uses labeled thymidine that is added to the cell culture during the final days of analysis. Cultured cells are exposed to levels of test drug higher than the peak levels achieved clinically, so-called extreme drug testing. If the percentage of surviving cells is greater than 1 standard deviation of the median survival seen in reference populations, then the cells are said to express extreme drug resistance (EDR). The significance of EDR to a particular compound equates to a less than 1% chance of a clinical response to the tested drug.¹³

Another commercially available product reports both drug sensitivity and resistance as a function of the dose-response curve determined for each specimen after exposure to a range of drug concentrations. After incubation with the drug of interest, viable cells are fixed and stained, then compared with controls. Assay validation uses comparison with ATP-based tests and relevant cell lines in the National Cancer Institute panel of 60 (NCI-60), and their response to the same drugs. Results of this assay have been reported to correlate with progression-free survival (PFS) in EOC patients.¹⁴

Evidence Supporting the Use of CSRAs

Retrospective studies comprise the majority of the literature concerning the use of CSRAs in EOC patients. Taylor and colleagues collected tumor samples or ascitic fluid from 120 patients prior to their primary treatment.

Chemotherapy sensitivity was determined after exposure to relevant drugs and cell survival was estimated using the MTT assay. The authors reported a significant correlation of in vitro sensitivity with clinical response, with an 85% PPV. They also noted that patients with sensitive tumors had significantly improved 5-year survival rates (24% vs 12% for the insensitive group; $P = .03$).¹⁵ However, this investigation has limited clinical utility in that a majority of patients are known to respond favorably in the primary setting (ie, high prevalence of chemosensitivity).

Conversely, Holloway and associates reported the use of a resistance assay in the primary treatment setting.¹⁶ This retrospective study examined 79 patients and demonstrated a significant improvement in PFS among those who had low levels of platinum resistance (24 months) compared with those with extreme platinum resistance (6 months; $P < .01$). Based on the accepted clinical definition of platinum resistance, the NPV of the assay was noted to be 47%. Again, these results would be expected in the primary setting, and this information would not result in the exclusion of a platinum agent in primary therapy—arguably, the most significant impact a test of this nature could have. However, an interesting finding in the study by Holloway is that patients with an in vivo resistance to taxane compounds were treated with platinum plus a nontaxane, usually cyclophosphamide (CP) or cyclophosphamide and adriamycin (CAP). Notably, the survival outcomes for patients who received assay-directed CP or CAP mirrored those who received the standard platinum + taxane combination. This implies that there could be a subset of patients for whom a platinum combined with an agent other than a taxane may offer better results than if they received standard platinum taxane therapy.

The reliable detection of this smaller population of patients would be clinically relevant if confirmed in a controlled study.

Gallion and colleagues retrospectively examined 135 cases of primary ($n = 84$, 62%) and recurrent ($n = 51$, 38%) ovarian cancer in which the patient received assay-directed therapy.¹⁴ The dose-response curves generated by the assay were categorized as: resistant (no detectable response), intermediate (35% cell kill at the highest dose), and sensitive (35% cell kill at the intermediate dose). They found that the assay was able to correctly predict a nearly 3 times increased risk of disease progression among tumors classified as resistant, compared with sensitive tumors. This study did not assess OS as an endpoint, which limits its clinical utility.

A more recent retrospective study examined 253 cases of primary ovarian cancer that had tissue samples submitted for EDR assay testing at the time of primary surgery.¹⁷ The authors examined EDR results for over 13 compounds tested. Dividing the compounds into 2 groups: standard therapy (platinum and taxane) and secondary therapy (other compounds), the authors sought to determine if EDR to both categories of compounds (ie, dual resistance) was associated with treatment response and OS. They found that dual resistance was not predictive of either clinical response to chemotherapy or OS. Only 28.5% of the cohort demonstrated no drug resistance, whereas a majority (89.5%) had EDR in up to 3 tested compounds *de novo*. Notably, the percentage of patients responding to chemotherapy was not affected by increasing numbers of resistant compounds. Even in 3 patients found to have EDR to 6 tested compounds, all responded to standard chemotherapy. In this, the largest retrospective study of a chemoresistance assay, there was no clinically useful information

gained from resistance testing in the primary setting.

A case-control investigation compared the recurrences of 50 patients with EDR assay-directed therapy with 50 matched controls treated empirically.¹⁸ Platinum-sensitive tumors had an improved overall response rate (ORR) (65% vs 35%; $P = .02$) and OS (38 months vs 21 months; $P < .01$) when treated with assay-directed therapy, compared with controls. In the platinum-resistant group, there was no demonstrated benefit to chemotherapy-resistance testing. Once again, platinum-sensitive patients are more likely to respond to chemotherapy compared with platinum-resistant patients, so these findings would not change clinical practice. Additionally, as the EDR assay used in this study has a reported 99% NPV,¹³ its utility in the setting of platinum resistance is low. These patients are likely to have tumors resistant to many drugs. Determining what might work, rather than what does not work, would be more clinically useful.

One prospective, randomized, controlled trial of chemotherapy sensitivity assay-directed therapy compared with empirical treatment in recurrent EOC was reported by Cree and colleagues in 2007.¹⁹ They used a luciferase-based assay that is not commercially available. Ninety-four (52%) patients were assigned assay-directed therapy, and 86 (48%) were treated according to their physician's choice. The study was powered to find a 20% difference in ORR among those patients receiving assay-directed therapy (assuming a baseline 30% response rate). Primary endpoints were PFS and ORR. Median follow-up was 18 months and results were analyzed by intent to treat. The authors found no significant differences between the 2 groups with respect to PFS, ORR, or OS. They did note a trend toward improved PFS. As the only randomized

trial to evaluate a chemotherapy sensitivity assay to date, this study demonstrates that assay-directed therapy is at least as good as empirical therapy. Larger prospective studies are needed to establish if there is truly an actual benefit of using this assay.

Limitations of CSRAs

Several of the major limitations of CSRAs stem from the need, in all cases, to use in vitro cell culture. As our understanding of the tumor microenvironment evolves, it is understandable that the artificial propagation of tumor cells promotes actions that may not mirror the behavior of cells in vivo. Cell culture is known to place unique stress on harvested cells, and the removal of supportive stromal cells, along with the associated growth signals, may impose further environmental pressure.²⁰ The genetic variations suited to survival in culture may yield an altered phenotype. Additionally, the immune system is known to interact with, and in some instances alter, the growth of tumor

cells.²¹ The authors demonstrated that 18.6% of primary and 26.1% of recurrent tumors possessed differences in EDR testing within the same patient. Recurrent tumors exhibited greater heterogeneity and more frequent EDR. The authors noted that, based on these data, one-fifth of primary tumors and over one-quarter of recurrent tumors are potentially undertreated. Although the EDR assay may alert us to tumor heterogeneity, the decision of which combination of chemotherapies to use is still a matter of clinical judgment.

The biology of the chemotherapeutic agents being tested is largely ignored in CSRAs. Tumors in vivo are exposed to the native agent and a variety of metabolites, some of which have significant activity depending on the drug being considered. CSRAs cannot account for the pharmacokinetics (what the body does to the drug) or pharmacodynamics (what the drug does to the body)—both of which influence the clinical tumor

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cells. The absence of active immune cells in cell culture, combined with phenotypic alterations as just described, may together result in significant survival bias after drug exposure.

The inherent heterogeneity present in nearly all tumors further complicates testing that is dependent on single tumor samples. It has been established that metastatic ovarian cancer implants have genetic differences when compared with the primary tumor. A recent study by McAlpine and associates evaluated, utilizing an EDR assay, multiple synchronous tumor samples in primary and recur-

rent ovarian cancer patients.²¹ The authors demonstrated that 18.6% of primary and 26.1% of recurrent tumors possessed differences in EDR testing within the same patient. Recurrent tumors exhibited greater heterogeneity and more frequent EDR. The authors noted that, based on these data, one-fifth of primary tumors and over one-quarter of recurrent tumors are potentially undertreated. Although the EDR assay may alert us to tumor heterogeneity, the decision of which combination of chemotherapies to use is still a matter of clinical judgment.

The American Society of Clinical Oncology (ASCO) last updated their Technology Assessment of CSRAs in 2004.²² Although this assessment has been criticized,²³ it does provide a framework for the evaluation of these tests. Citing the lack of prospective information, the working group did not find sufficient evidence to recommend the routine use of CSRAs.

Gene-Based Chemotherapy Response Prediction

The next generation of chemotherapy response testing will account for several of the limitations inherent in CSRAs. Gene-based assays are not dependent on the establishment of clonally expanded tumors, and are not limited by the effects of the tumor microenvironment or drug metabolism. Gene expression profiling, through the use of microarray data, can be used to link 2 cell populations—and in turn, use the reference population to predict the chemotherapy response of the test population.²⁴ Lee and colleagues evaluated gene expression profiles from a reference set of untreated cancer cells (the NCI-60 panel) that have known in vitro sensitivity to specific agents. Using a statistical model, the researchers found a predictive cohort of genes and then used them to form a gene expression model (GEM). Evaluation of this GEM on a target set of tumors provided the probability of clinical response. This

method has successfully predicted the clinical outcome in several bladder and ovarian cancer clinical studies where patients had gene expression profiling prior to treatment with chemotherapy.^{24,25}

Conclusions

In vitro chemotherapy sensitivity and resistance testing may still hold promise for clinical decision-making, improved survival, and limiting unnecessary toxicity in the treatment of EOC. Given the current limitations and lack of randomized, controlled results, these assays are best used in the setting of a clinical trial. ■

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Main Points

- Epithelial ovarian cancer (EOC) claims the lives of more US women annually than any other gynecologic malignancy. Poor prognosis observed with EOC relates to the fact that a majority of women present with late-stage disease, and tumors that initially respond to chemotherapy inevitably manifest resistance.
- Phenotypic categorization of tumors in general, and EOC specifically, has the potential to enhance chemotherapy response rates. Several commercially available tests, known as chemotherapy sensitivity and resistance assays (CSRAs), seek to predict a clinical tumor response based on in vitro cell behavior. The concept aims to test each particular patient's tumor and determine the specific chemotherapy that will provide maximum response.
- Chemotherapy sensitivity assays predict agents, or combinations of agents, to which a particular tumor may respond. Chemotherapy resistance assays identify agents to which a particular tumor may not respond (ie, sensitivity assays more effectively establish the positive predictive value and resistance assays more effectively establish the negative predictive value).
- Several major limitations of CSRAs stem from the need to use in vitro cell culture. Cell culture is known to place unique stress on harvested cells, and the removal of supportive stromal cells, along with the associated growth signals, may impose further environmental pressure. Genetic variations suited to survival in culture may yield an altered phenotype.
- Gene-based assays are not dependent on the establishment of clonally expanded tumors, and are not limited by the effects of the tumor microenvironment or drug metabolism. Gene expression profiling, through the use of microarray data, can be used to link 2 cell populations—and in turn, use the reference population to predict the chemotherapy response of the test population.
- In vitro chemotherapy sensitivity and resistance testing may still hold promise for clinical decision making, improved survival, and limiting unnecessary toxicity in the treatment of EOC. Given the current limitations and lack of randomized, controlled results, these assays are best used in the setting of a clinical trial.

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